The AI Robot Race and Its Hidden Cost

A 10-Year Outlook (2025–2035) — USA vs. China

Sandy E. Quintero

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Abstract

Between 2025 and 2035 the United States and China are racing to build the humanoid robot, and the contest is usually scored by a single question: who finishes first. This book argues that this is the wrong question. Drawing only on primary and authoritative sources — bank research desks, government filings, UN and industry statistics, peer-reviewed economics — it follows one machine across its entire lifecycle, from the bill of materials that builds it to the waste stream that swallows it, and prices each stage in turn.

What emerges is an asymmetry that no single stage reveals on its own. The race is extraordinary at *acquiring* the machine and very nearly blind to *keeping* it. Capital, materials, manufacturing, and deployment — the front half of the lifecycle — are optimized with national urgency: a China-built bill of materials engineered to fall from about $35,000 today to under $17,000 by 2030 (Bank of America Institute, 2026), a roughly $138 billion state fund (International Federation of Robotics, 2025a), a $39 billion private valuation for a company yet to sell at scale (Figure AI, 2025), and export controls and tariffs fought over the magnets and chips inside every robot (Congressional Research Service, 2025; International Energy Agency, 2025; Office of the United States Trade Representative, 2024). The back half — reliability, obsolescence, disposal, and recovery — is barely measured at all: no manufacturer discloses a humanoid's mean time between failures (Bain & Company, 2025), no robot-specific e-waste tonnage exists, and recycling supplies only about 1% of the world's rare-earth demand even as nations fight to secure that same material at the mine (Baldé et al., 2024).

That gap is the hidden cost of the title. It is not a price the market has set too low; it is a half of the machine's life the race has chosen not to measure at all — and the silence in the data falls exactly where the spending and the urgency do not. Because that is a matter of where attention and money are pointed, not of any technical barrier, the book's closing argument is that the gap is a default, not a destiny: a humanoid built in our own image reflects the priorities we build into it, and which half of its life we choose to value is still an open question.

Key findings

The race optimizes the four *acquisition* stages of a humanoid's life and neglects the four *keeping* stages. Every figure below is verified against its primary source in the chapter indicated.

The acquisition half — optimized:

- **Build (Ch. 2):** A China-built bill of materials runs about $35,000 in 2025 and is engineered to fall below $17,000 by 2030; actuators and dexterous hands are roughly 70% of that bill, while onboard compute — the "brain" — is only about 10% (Bank of America Institute, 2026).
- **Fund (Ch. 3):** China stood up a ¥1 trillion (~$138 billion) state "hard-tech" fund (International Federation of Robotics, 2025a); in the United States, Figure AI alone reached a $39 billion private valuation before selling robots at scale (Figure AI, 2025).
- **Source (Ch. 5):** China refines ~91% of the world's rare earths and makes ~94% of its permanent magnets (International Energy Agency, 2025); the United States controls the frontier chips and now taxes the magnets it barely makes at 25% (Congressional Research Service, 2025; Office of the United States Trade Representative, 2024).
- **Deploy (Ch. 6):** Robot density has reached 1,012 per 10,000 workers in South Korea and 470 in China — past the United States at 295 — against a world average of 162, pulled by aging workforces (Acemoglu & Restrepo, 2022; International Federation of Robotics, 2024a).

The keeping half — neglected:

- **Run (Ch. 4):** No humanoid mean-time-between-failures has ever been disclosed, and today's machines run only about two hours on a charge (Bain & Company, 2025).
- **Outdate (Ch. 7):** The brain is lapped about every twelve months while the body is built to last a decade; Amazon shortened the booked life of its servers from six years to five and named AI as the reason (Amazon.com, Inc., 2025a; NVIDIA, 2024a; The Robot Report, 2019).
- **Discard (Ch. 8):** Global e-waste is rising from 62 to 82 million tonnes while the share formally recycled falls toward 20%; only about 1% of rare-earth demand is met by recycling (Baldé et al., 2024).

- **Recover (Ch. 9):** ABB's remanufacturing loop cuts CO_2 by 75% and reuses 60–80% of an industrial arm — real, but a rounding error against 4.66 million robots in service, and it does not transfer to the fast-obsoleting humanoid (ABB, 2026; International Federation of Robotics, 2025b).

A note on method

Every figure in this book passed through a single discipline: a claim is either traced to a primary or authoritative source or it is not used. Bank research was read from source PDFs where machine-readable; government filings (SEC 10-K and 10-Q filings, the USTR Section 301 determination) and UN, IEA, IFR, and Pew data were taken from the primary documents. Where sources conflicted, the book reports a sourced range rather than an invented midpoint. Where a widely circulated figure — a "200–500 hour humanoid MTBF," a "one billion robots by 2050" e-waste projection, a generational disposal-behavior matrix — could be traced only to unsourced drafts, it was discarded and the absence stated plainly. In several places across the keeping half, the honest finding is that the data does not yet exist; that absence is itself reported as evidence.

1. The Superpower Duel

The race to build humanoid robots is not one contest but two, run on different tracks toward different finish lines. The United States and China have each bet on a distinct theory of how an industry like this is won, and the divergence in those bets explains almost everything that follows in this report.

The American wager is **software-first and capital-intensive**: win the brain, and the body will follow. The Chinese wager is **hardware-first and state-directed**: drive the cost of the body toward commodity pricing, and scale will follow. These are tendencies, not absolutes — Tesla and Boston Dynamics build formidable hardware, and Chinese firms train their own models — but the center of gravity of each program is unmistakable, and neither bet is obviously wrong. The question this report exists to answer is which theory survives contact with the *hidden cost* of building at scale; but first we have to see the two machines clearly.

1.1 The American model: betting on the brain

The defining American move has been to treat the humanoid as an AI problem wearing a chassis. The clearest signal is NVIDIA's Project GR00T, a general-purpose foundation model for humanoid robots launched in March 2024 and paired with simulation tools that let a robot learn skills from synthetic data rather than from slow, expensive physical trials (NVIDIA, 2024b). The reasoning is the reasoning of frontier AI: if intelligence can be trained in the cloud and poured into any body, the body becomes a commodity and the *model* becomes the moat.

That conviction is priced into American capital markets. Figure AI, an American humanoid startup, raised $675 million in early 2024 at a reported $2.6 billion valuation, backed by Microsoft, the OpenAI Startup Fund, NVIDIA, and Jeff Bezos, alongside a collaboration with OpenAI to give its robots language and reasoning (Figure AI, 2024). By September 2025 the same company had closed a Series C of more than $1 billion at a **$39 billion** post-money valuation (Figure AI, 2025). These are software-industry numbers — valuations that assume the winner captures a platform, not a product line.

The cost philosophy follows from the capital philosophy. At Tesla's October 2024 *We, Robot* event, Elon Musk put Optimus's eventual price at $20,000–$30,000 — "less than a car" — though that is a long-range target for a humanoid Tesla has not yet sold (TechCrunch, 2024). Agility Robotics' Digit, by contrast — offered largely through leasing rather than outright sale — has been reported at upwards of $250,000 per unit (The Robot Report, 2024). The American bet tolerates a high unit cost today on the assumption that scaled intelligence will collapse it tomorrow.

1.2 The Chinese model: betting on the body

China inverted the bet. Rather than chase the frontier model, it set out to make the *physical robot* cheap, abundant, and domestically supplied — and it did so from the top down. In November 2023 the Ministry of Industry and Information Technology issued national guidance naming humanoid robots a disruptive technology on the order of the computer, with targets to reach mass production "at an internationally advanced level" by 2025 and to make humanoids "an important new engine of economic growth" by 2027 (South China Morning Post, 2023; U.S.-China Economic and Security Review Commission, 2024).

This is not exhortation without money behind it. At the 2025 Two Sessions, the National Development and Reform Commission announced a ¥1 trillion (about **$138 billion**) state fund for "hard tech" — robotics and AI among them — deployed as market-based equity with horizons of up to twenty years (International Federation of Robotics, 2025a; The Robot Report, 2025). The capital is already moving: by mid-2025, on The Robot Report's tally, China had put an estimated $3.4 billion into new robotics ventures — 42% more than the United States and five times Europe — while six of its eleven major humanoid manufacturers had launched mass-production lines in 2024 (The Robot Report, 2025).

China's structural advantage, though, is not the subsidy — it is the supply chain it already owns. A decade of electric-vehicle dominance built much of the component base a humanoid needs, and nowhere is the edge clearer than in batteries. In 2024, lithium-ion battery pack prices averaged **$94/kWh in China** — the world's lowest — against a global average of $115/kWh, with US and European packs running 31%

and 48% higher *than China's* (BloombergNEF, 2024). The same battery supply chain that made Chinese EVs cheap now feeds Chinese robots. The emblem is Unitree's G1: the standard model launched in 2024 at about **$16,000** and Unitree now lists that same base configuration at **$13,500** (The Robot Report, 2024; Unitree Robotics, 2026). The comparison demands an honest caveat — at 1.32 m and roughly 35 kg, the G1 is a smaller, lighter machine than the full-size Optimus or Digit, so it is not a like-for-like substitute (Unitree Robotics, 2026). Yet the gap is far too wide to be explained by size alone: China is *shipping* a capable humanoid for the price of a used car today, while Tesla still only *targets* $20,000–$30,000 for an Optimus it has not sold (TechCrunch, 2024) and Agility's full-size Digit is reported at upwards of $250,000 (The Robot Report, 2024).

> The United States is racing to build the *smartest* robot; China is racing to build the *cheapest* one. Whether intelligence or affordability turns out to be the binding constraint of the next decade is the question that decides the race.

1.3 Two theories, one table

Dimension	United States	China
Core bet	Software-first: the foundation model is the moat	Hardware-first: the cheap, domestic body is the moat
Engine of progress	Private capital + frontier AI	State direction + industrial supply chain
Emblematic asset	NVIDIA GR00T foundation model (NVIDIA, 2024b)	Unitree G1, now $13,500 (Unitree Robotics, 2026)
Capital mobilization	Figure AI: $39B private valuation (Figure AI, 2025)	NDRC: ¥1T (~$138B) state hard-tech fund (International Federation of Robotics, 2025a)
Cost posture	High unit cost now; betting scale collapses it	Low unit cost now, via EV-derived parts (battery $94/kWh) (BloombergNEF, 2024)

Note: the two "capital mobilization" entries are not the same unit — one firm's private valuation versus a national fund — but each is the clearest emblem of how money is marshaled on its side.

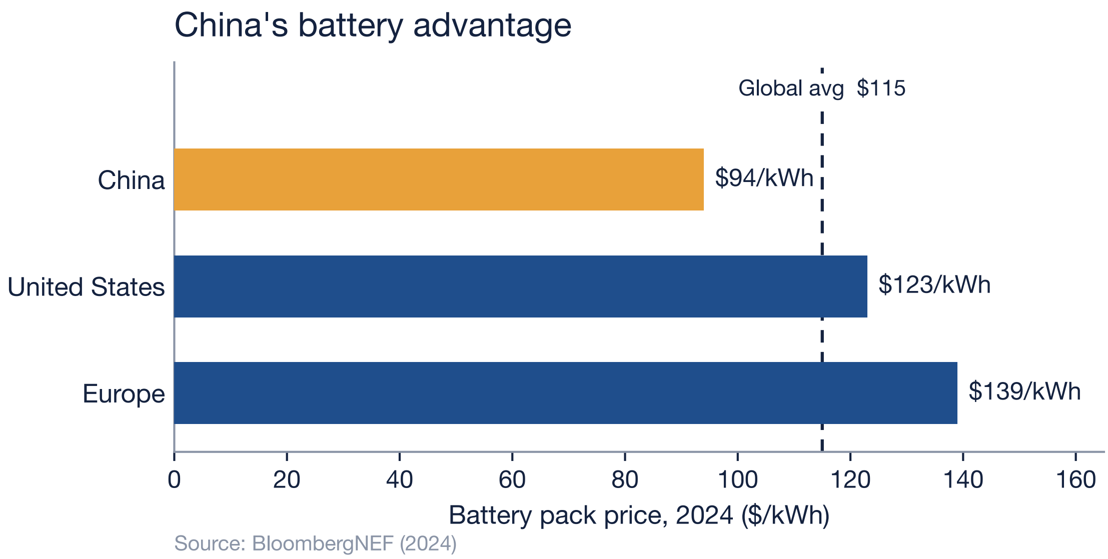

Figure 1.1: Lithium-ion battery pack prices by region, 2024 ($/kWh). China's $94 packs are the world's cheapest; the United States ($123) and Europe ($139) run 31% and 48% higher — both derived from BNEF's reported percentages off China's base. The dashed line marks the $115 global average. Source: BloombergNEF (2024).

These two philosophies are not merely strategies; they are the opening conditions of the robot's entire life. Everything a humanoid will cost — to build, to run, to repair, and eventually to discard — is set in motion by which of these models produced it. The next chapter opens that machine up and counts the cost, component by component.

2. The Economics of a Humanoid

Strip away the strategy and a humanoid robot is, at the moment of its birth, a list — a bill of materials naming every motor, bearing, sensor, cell, and screw that has to be bought and bolted together before the machine can stand, walk, or close a hand around a cup. That list is where the two philosophies of Chapter 1 stop being theories and start being invoices. It is also where one of the report's fastest-moving numbers lives: the cost of building one of these machines is at once very high and falling very quickly.

In early 2024 Goldman Sachs reported that the manufacturing cost of a humanoid had fallen from a range of $50,000–$250,000 per unit to $30,000–$150,000 — the spread running from a basic model to a state-of-the-art one — a roughly 40% drop in a single year, against the 15–20% annual decline the bank had expected (Goldman Sachs, 2024). Whatever else this report concludes, it has to be read against that slope: the economics that will decide the race are not a fixed obstacle but a collapsing one, and the question is which side of the race the collapse favors.

2.1 Opening the machine: the cost is mostly motion

Open the chassis and the money is concentrated in one place — movement. By 2030, Bank of America Global Research projects that linear actuators will be the single largest line in the bill of materials at 27%, with rotary actuators close behind at 24%; the two actuator modules alone are more than half the bill, and the equally movement-driven dexterous hands add a further 19%, so the parts that simply make the robot *move* come to fully 70% (Bank of America Institute, 2026). Morgan Stanley, working from its own estimates, lands in the same place: actuator modules represent "just under half" of a humanoid's total cost (CNBC, 2026). Two of the most-watched research desks on Wall Street independently agree that the most expensive thing about a humanoid robot is not its mind but its muscles.

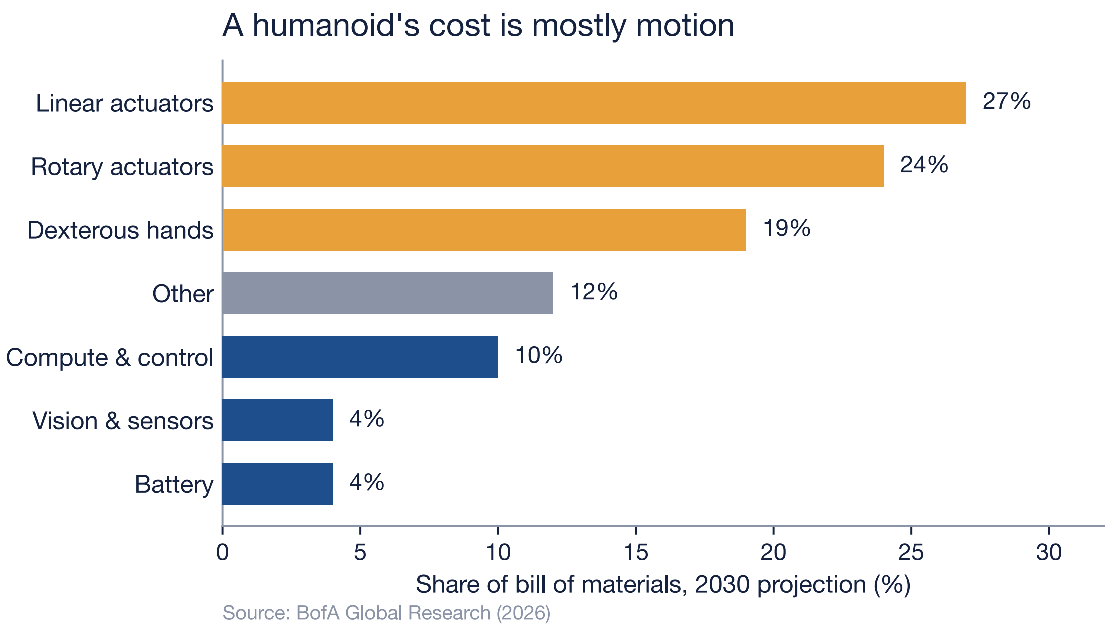

Figure 2.1: Where the money goes in a humanoid robot — BofA Global Research's projected 2030 bill of materials, by module. The two actuator modules (linear and rotary) are 51% of the bill on their own; adding the equally movement-driven dexterous hands, everything that moves the robot reaches 70% (amber), while the onboard 'brain' of compute and control is just 10%. Source: BofA Global Research (2026).

This is the quiet rebuttal to the American wager of Chapter 1. If the foundation model is the moat, the bill of materials says the moat is the cheap part: onboard compute and control come to only about 10% of the projected 2030 bill, while the actuators, precision reducers, and articulated hands that let a robot *act* in the physical world dominate it (Bank of America Institute, 2026). Intelligence, in other words, is becoming the cheap part of a humanoid — at least to reproduce. A foundation model is enormously expensive to *train*, but that is a fixed cost spread across every unit it ships in, and following it is the work of the next chapter; each additional body, by contrast, has to be built again from machined metal and precision supply chains, not poured from a finished training run.

> A humanoid's brain is the cheap part. Its body is the bottleneck — and bodies are won in factories, not in data centers.

One number in that breakdown deserves a second look, because it quietly overturns a figure the early drafts of this research took for granted. Batteries are projected to be just **4%** of the 2030 bill of materials (Bank of America Institute, 2026), not the 15–20% that older estimates assumed. The reason is the collapse in cell prices documented in Chapter 1: at roughly $94/kWh in China, the energy system has fallen from a major cost center to a rounding error (BloombergNEF, 2024). The same EV-derived battery supply chain that gave China its edge has, across the whole industry, made the battery the least of a robot-builder's problems.

2.2 Why "the cost" is four different numbers

Ask what a humanoid costs and the published answers scatter from $35,000 to $200,000 — a spread wide enough to make any single figure look invented. The early drafts feeding this report did exactly that, asserting a US build cost of $71,000 against a Chinese $46,150 in one table and far lower numbers in another, with no source either could be traced to. The scatter is not contradiction; it is four honest measurements of four different things, and once they are sorted the picture is coherent.

Published "cost"	What it actually measures	Figure	Source
China-built BOM, 2025	Parts only, on a mature supply chain	~$35,000	BofA (Bank of America Institute, 2026)
Manufacturing range, 2024	Per unit, basic to state-of-the-art	$30,000–$150,000	Goldman Sachs (Goldman Sachs, 2024)
Pilot-stage development	Parts, before mass production	$90,000–$100,000	BofA (Bank of America Institute, 2026)
Fully loaded, high-income economy, 2024	Total cost of one finished unit	~$200,000	Morgan Stanley (Morgan Stanley, 2025)

A bill of materials built on China's mature component base runs about $35,000 today and is expected to fall below $17,000 by 2030 as scale and component redesign compound (Bank of America Institute, 2026). A robot still in pilot-stage development, before any of that scale arrives, costs $90,000–$100,000 in parts alone (Bank of America Institute, 2026). And the *fully loaded* cost of one finished unit in a high-wage economy — parts, assembly, and everything around them — was around $200,000 in 2024 on Morgan Stanley's accounting, on a path to roughly $150,000 by 2028 and, in the bank's longer-range view, $50,000 by 2050 (Morgan Stanley, 2025). The drafts' $71,000 and $46,150 were not wrong so much as unanchored: points floating inside Goldman's sourced $30,000–$150,000 band with no stated year, model class, or boundary around what was being counted.

2.3 The geography of a bill of materials

Where a humanoid is built changes its birth cost as much as what it is, and the reason traces straight back to Chapter 1. Morgan Stanley estimates that automotive-parts suppliers can account for close to 60% of a humanoid's production cost (CNBC, 2026). That is the EV supply chain reappearing as a robot supply chain: the actuators, reducers, cast frames, and battery packs that dominate the bill of materials are, to a large degree, the components a mature car industry already knows how to make at volume. China's advantage in the body of the robot is the same advantage it built in the body of the car, and it is structural, not merely subsidized. It is also why China's mature base yields the cheapest bill of materials in that earlier table — a pre-scale pilot build costs nearly three times as much in parts alone — for the simplest of reasons: the lowest bill of materials belongs to whoever already runs the supply chain at volume (Bank of America Institute, 2026).

The bill of materials, though, is only the price of being born. It says nothing about who pays it — and on that question the two superpowers diverge as sharply as they do on everything else. In the United States the build cost is fronted by private capital betting on a platform; in China it is underwritten by the state betting on an industry. The next chapter follows the money.

3. Capital & Investment Landscape

Chapter 2 ended on a question the bill of materials cannot answer: a humanoid costs tens of thousands of dollars to build before it earns a cent, so *who fronts that money* — and on what terms — decides who gets to build at all. Here the two strategies of Chapter 1 stop being philosophies of engineering and become philosophies of finance. The United States funds its robots the way it funds software: private capital, concentrated into a few presumed winners at valuations that assume a platform. China funds its robots the way it funds infrastructure: state-directed capital, spread across an entire industry on a horizon no venture fund could tolerate. The machines may converge; the money behind them could not be more different.

3.1 The American model: private capital, betting on platforms

American humanoid robotics is financed by the deepest private-capital market on earth, and it behaves accordingly. Venture funding into robotics startups surged to a multiyear high in 2025, and the single largest humanoid rounds on record were American (Crunchbase News, 2025). What is striking is not the aggregate but its *concentration*. Figure AI, which has yet to sell a robot at scale, closed a Series C above $1 billion in September 2025 at a **$39 billion** post-money valuation (Figure AI, 2025). Physical Intelligence, a startup building robot foundation models, was valued at **$5.6 billion** in November 2025 after a $600 million round, more than doubling from $2.4 billion a year earlier, with Jeff Bezos, the OpenAI Startup Fund, Thrive Capital, and Lux Capital among its backers (Bloomberg, 2025). Apptronik, maker of the Apollo humanoid, pushed its Series A past $935 million at roughly a **$5.5 billion** valuation, drawing in B Capital and Google (Crunchbase News, 2026).

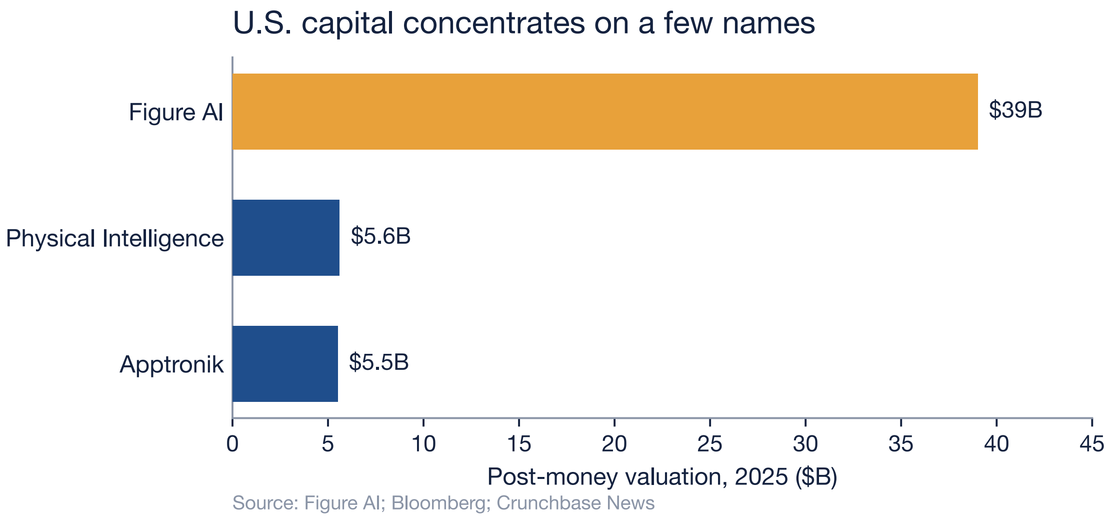

Figure 3.1: Private valuations of leading U.S. humanoid-robot startups (post-money, USD billions). Figure's $39B (amber) — for a company yet to sell a robot at scale — dwarfs even its well-funded peers, the signature of a capital model that concentrates enormous bets on a few presumed winners. Sources: Figure AI (2025); Bloomberg (2025); Crunchbase News (2026).

These are software valuations attached to hardware companies, and they reveal the logic of the bet. Private capital is not trying to fund an industry; it is trying to *own the platform* that wins it, the way earlier funds tried to own the search engine or the social network. The money arrives fast and in enormous concentrations — from venture syndicates and from the balance sheets and venture arms of NVIDIA, Microsoft, OpenAI, and Google — but it arrives expecting an exit, on the seven-to-ten-year clock of a fund's life. It rewards the company that looks most like the eventual monopolist and starves the rest. The strength of the model is its speed and its appetite for moonshots; its weakness is that it concentrates the nation's bet on a few names and demands they pay off before the fund closes.

3.2 The Chinese model: state-directed capital, betting on an industry

China's capital arrives from the opposite direction — top-down, patient, and diffuse. At its apex sits the national venture-capital guidance fund the National Development and Reform Commission announced in 2025: about ¥1 trillion (**$138 billion**), deployed as market-based equity over a twenty-year horizon, with each regional sub-fund expected to exceed ¥50 billion (International Federation of Robotics, 2025a). Beneath it runs a second national vehicle, an $8.2 billion AI Industry Investment Fund launched in January 2025 to steer capital into frontier technologies including embodied AI (Jamestown Foundation, 2025). And beneath *that* lies a dense lattice of local money: dedicated robotics or humanoid funds of roughly ¥10 billion each stood up by Beijing, Shanghai's Pudong district, and Shenzhen, with local governments committing more than ¥70 billion to humanoid robotics in total (TMTPost, 2025), alongside provincial subsidies that reach up to ¥100 million per approved project in Guangdong and ¥200 million for research institutions in Suzhou (Jamestown Foundation, 2025).

This is what Chinese policy literature calls a "whole-of-nation" effort, and its private market reflects it: financing into China's humanoid sector exceeded ¥10 billion (about $1.4 billion) in the first half of 2025 alone, a record, much of it state-linked rather than purely private (Jamestown Foundation, 2025). The strength of the model is patience and breadth — capital that can wait twenty years and seed a whole supply chain rather than a single champion. Its weakness is the mirror image: state money flowing to a crowded field invites duplication and misallocation, dozens of near-identical firms chasing the same subsidies, and a reckoning when priorities shift. Where American capital risks betting too narrowly, Chinese capital risks betting too widely.

3.3 Two ledgers, one race

Dimension	United States	China
Capital source	Private VC + tech-giant balance sheets	State guidance funds + local government
Emblematic vehicle	Figure AI: $39B private valuation (Figure AI, 2025)	NDRC: ¥1T (~$138B) 20-year fund (International Federation of Robotics, 2025a)
Logic of the bet	Own the winning platform	Build the whole industry
Time horizon	A fund's life (~7–10 years), exit-driven	Up to 20 years, strategy-driven
Characteristic failure	Over-concentration in a few names	Over-subsidized, crowded field

Note: the two emblematic vehicles are not the same unit — one firm's private valuation versus a national fund — but each is the clearest signature of how capital is raised on its side, exactly as in Chapter 1's strategic comparison.

The asymmetry that matters most is not the size of the two pools but their *patience.* American capital must be returned; the fund that backed Figure at $39 billion needs an exit, which means the robot has to become a profitable product on a venture timeline. Chinese state capital carries no such clock — it can absorb a decade of losses if the result is a domestic industry that no longer depends on foreign suppliers. That difference will reappear, transformed, when this report reaches the *hidden* cost: a financing model that demands quick returns has every incentive to externalize the long-tail costs of repair, obsolescence, and disposal onto someone else — and, as later chapters will show, a state racing for industrial dominance externalizes them no less freely, onto its own soil and rivers. Neither model has yet priced what the machine leaves behind. Capital can build the machines and price their parts; whether the machines can *stay built* — whether they survive contact with the physical world long enough to repay anyone — is the question the next chapter opens.

4. The Reliability Gap

Chapter 3 left a humanoid built, priced, and funded — but a robot that has been paid for has only reached the starting line. The test that matters comes next, in the *working life*: can the machine show up, day after day, and do the job without breaking? Industrial robots passed that test decades ago. Humanoids have not been running long enough for anyone to honestly say whether they can pass it at all — and that absence of evidence, as much as any number, is the finding of this chapter.

4.1 The proven machine

The industrial robot arm is one of the most reliable machines ever mass-produced, and it has the operating history to prove it. Manufacturers advertise mean-time-between-failures figures as high as 100,000 hours — more than eleven years of continuous running — and while independent researchers treat those marketing numbers with open skepticism, the skepticism cuts in an instructive direction (The Robot Report, 2019). A study spanning more than 400 factories, published in the *International Journal of Performability Engineering* and reported by The Robot Report, found that robot *cells* achieve about 88% availability — real downtime, not the advertised perfection — but that **80% of stoppages have nothing to do with the robot at all**, tracing instead to the conveyors, sensors, grippers, and other peripherals around it (The Robot Report, 2019). The robot is implicated in only the remaining fifth — about **2.5% of total available time** — and of *that* slice, only a quarter is the robot physically breaking; most of the rest are positional faults (The Robot Report, 2019). The headline MTBF may be marketing, but the underlying truth survives the debunking: the arm itself is almost never the thing that breaks. It is a solved problem, and it is solved because the machine is rigid, repetitive, bolted to a floor, and asked to do one motion a few million times.

4.2 The unproven machine

A humanoid is none of those things, and its reliability record reflects it — which is to say, there is barely a record at all. The most basic limit is the most telling, and it is not the one Chapter 2 priced: the battery may be cheap to buy — about 4% of the bill of materials in BofA's 2030 projection — but it is still short, and most humanoid robots today operate for only about **two hours** on a single charge (Bain & Company, 2025). Bain & Company projects that better batteries could extend that to roughly six hours by 2030, but that a full eight-hour shift "could remain elusive" for a decade or more (Bain & Company, 2025). A machine that must stop to recharge three or four times a day has not yet earned the word *reliable*; it has not even reached the point where mean-time-between-*failures* is the binding constraint, because mean-time-between-*recharges* stops it first.

Behind that limit sits a deeper one: no manufacturer publicly discloses a humanoid MTBF, and Bain & Company's 2025 survey of the field cites no reliability or total-cost-of-ownership figures at scale at all (Bain & Company, 2025). Most deployments remain pilots, still leaning on human input for navigation, dexterity, and task-switching, and Bain cautions that polished demonstrations "often mask technical constraints through staged environments or remote supervision" (Bain & Company, 2025). The reliability has not been measured because the machine has not yet run long enough, in enough places, to measure it. The figures that circulate for humanoid service intervals — every few hundred operating hours — trace not to fleet data but to analysts' models; this report does not repeat them as fact, because none can be sourced to a machine that has actually logged the hours.

4.3 A gap measured against an absence

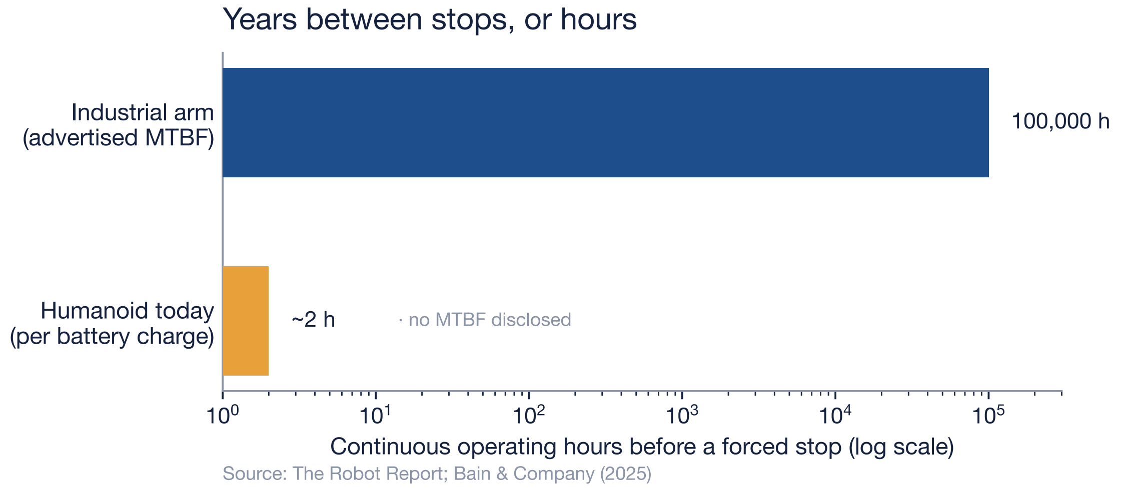

Figure 4.1: The reliability gap, on a logarithmic scale of continuous operating hours before the machine must stop. The industrial arm's bar is its advertised mean time between failures (up to ~100,000 hours); the humanoid's bar is its battery runtime per charge (~2 hours, Bain & Company), because no manufacturer discloses a humanoid MTBF to plot. The two bars measure different kinds of stop — a failure versus a dead battery — and that is precisely the point: on the humanoid side, the failure data does not yet exist. Sources: The Robot Report; Bain & Company (2025).

Set side by side, the two machines are separated by nearly five orders of magnitude — the industrial arm runs for years between failures while the humanoid runs for hours between recharges — but the more honest way to read the chart is that *one bar is data and the other is a placeholder.* We can state the industrial arm's reliability because the world has measured it across hundreds of thousands of machine-years. We cannot state the humanoid's, because the machine-years do not exist, and the limit we can measure — the battery — is so short that the question of long-run failure has not yet had the chance to be asked. A reliability gap this wide is not a tuning problem to be closed in a product cycle; it is the difference between a tool and a prototype.

That gap is also a cost the earlier chapters did not price. A machine that stops every two hours, is serviced on a schedule no one can yet quantify, and may be replaced rather than repaired when a newer model ships, consumes parts and labor across its whole working life — and those parts come from a supply chain that, by 2025, had become a front line in the trade war between the two superpowers. The next chapter follows the components across that border.

5. Trade, Tariffs & Supply Chains

Chapter 4 ended with a machine that consumes parts across its whole working life — and those parts do not come from one country. A humanoid robot is, physically, a treaty between two superpowers that are no longer on speaking terms. Its body is built from materials China controls; its mind is trained on chips the United States controls. Neither side can build a complete robot without reaching across the very border it is busy fortifying. This chapter maps that border, and the tolls now charged to cross it.

5.1 The body: China's grip on the magnet

Every claim in Chapter 2 about where a humanoid's cost lives — actuators above half the bill of materials, the dexterous hands pushing the mechanical total toward 70% — points at one material. An electric actuator's torque comes from a permanent magnet, and the magnets strong enough to move a robot's limbs are made from neodymium-iron-boron, alloyed with the heavy rare earths dysprosium and terbium so they hold their strength when the motor runs hot. There is no like-for-like commercial substitute at that power density, and there is, for practical purposes, one supplier.

China refined about **91% of the world's rare earths in 2024 and produced roughly 94% of the world's sintered permanent magnets** — up from around half two decades earlier (International Energy Agency, 2025). The grip is not evenly distributed: China mines about 60% of the magnet rare earths but refines and manufactures far more, so its control deepens at exactly the step where raw ore becomes a finished motor component (International Energy Agency, 2025). A country can open a mine elsewhere in a few years; building the chemical separation and metallurgy to turn that ore into magnet-grade material is the part China spent decades and enormous environmental cost to master.

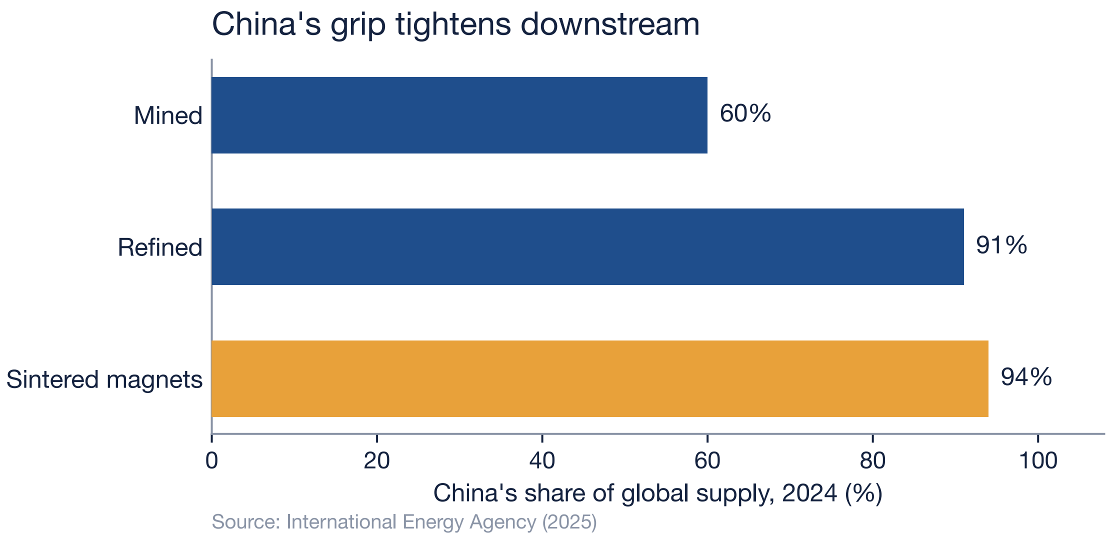

Figure 5.1: China's share of the rare-earth permanent-magnet supply chain, 2024. The grip tightens at every step downstream — from 60% of the magnet rare earths mined to 91% of refining to 94% of finished sintered magnets, the neodymium-iron-boron magnets inside almost every humanoid actuator. The amber bar is the chokepoint that matters most for robotics: the step nearest the finished robot is the step China controls most completely. Source: International Energy Agency (2025).

For most of those decades the dominance was commercial, not coercive. That changed in the open. On **4 April 2025, China placed seven medium and heavy rare earths — samarium, gadolinium, terbium, dysprosium, lutetium, scandium, and yttrium — together with their compounds, metals, and the magnets made from them, under an export-licensing regime** (Dezan Shira & Associates, 2025; International Energy Agency, 2025). Nothing was banned; every shipment now simply requires a license, an end-user certificate, and Beijing's permission, and the IEA recorded sharp declines in export volumes in the weeks that followed (International Energy Agency, 2025). A licensing queue is a quieter weapon than a ban — it throttles rather than severs, and it can be loosened or tightened by the day — but for a factory waiting on dysprosium it is the same empty loading dock. By October 2025 the controls had been extended to five more elements and, more strikingly, to *foreign-made* parts and assemblies that contain Chinese rare earths or were produced with Chinese technology — an attempt to follow the material across other countries' borders (International Energy Agency, 2025). That extraterritorial reach was fast-moving and partly suspended, so it is read here only as direction of travel; the April licensing regime is the settled fact.

Magnets are not the only chokehold. Two years earlier China had rehearsed the move on a smaller stage: in July 2023 it announced export licensing on **gallium and germanium**, the minor metals behind power semiconductors, fiber optics, and the infrared sensors used in some machine vision, effective that August (International Energy Agency, 2023). The 2023 measure was the proof of concept; the 2025 magnet controls were the same instrument aimed at the heart of the robot's body.

5.2 The brain: America's grip on compute

The United States holds the mirror image. A humanoid's intelligence — the foundation models of Chapter 1, trained on thousands of accelerators — depends on advanced chips and the equipment that makes them, and here the export controls run the other way. Beginning **7 October 2022, the Commerce Department's Bureau of Industry and Security restricted the export of advanced computing chips and the semiconductor-manufacturing equipment needed to produce them, and extended the Foreign Direct Product Rule** so the restrictions reach chips made anywhere in the world with U.S. technology; the rules were tightened again on 17 October 2023 (Congressional Research Service, 2025). Where China controls the materials that move a robot's body, America controls the silicon that trains its mind. The symmetry is almost too neat: one superpower owns the muscles, the other owns the brain, and a finished humanoid needs both.

5.3 The tariff paradox

Tariffs sit on top of the export controls as a second, blunter toll — and on the American side they collide with the chokepoint the country least controls. In the four-year review of its Section 301 duties, finalized in September 2024, the United States raised tariffs on a list of Chinese goods that reads like a humanoid's parts bin (Office of the United States Trade Representative, 2024).

Table 5.1: Final Section 301 tariff increases on Chinese imports, by product and effective year. Source: Office of the U.S. Trade Representative (2024).

Imported from China	New Section 301 tariff	Effective
Electric vehicles	100%	2024
Solar cells	50%	2024
Semiconductors	50%	2025
Lithium-ion EV batteries	25%	2024
Lithium-ion non-EV batteries	25%	2026
Permanent magnets	**25%**	**2026**
Natural graphite	25%	2026
Steel & aluminum products	25%	2024
Ship-to-shore cranes	25%	2024

Read Table 5.1 against the previous section and the paradox is plain. The United States has placed a 25% tariff on the **permanent magnets** it cannot yet make in volume, and on the **natural graphite** that lines a battery's anode — taxing the very imports its own factories depend on, because domestic production of either is only beginning to scale (Office of the United States Trade Representative, 2024). A tariff is meant to make a foreign good less attractive than a home-made one; here there is barely a home-made one to protect. The duty does not redirect the purchase so much as raise its price, and that surcharge flows downstream into the cost of every robot assembled on American soil. The 50% duty on semiconductors and the 100% wall around electric vehicles are coherent industrial policy; the magnet tariff is closer to a tax the buyer pays to itself for a dependency it has not escaped.

5.4 A machine that cannot be decoupled

Put the two arsenals side by side and the strategic picture resolves. Each superpower can raise the cost of the other's robots, and neither can yet build its own without the other's inputs. China can throttle the magnets in every actuator with a licensing pen; the United States can starve the training runs behind every robot's intelligence with the Foreign Direct Product Rule. The humanoid is the rare product that sits squarely in both lines of fire — body indexed to Beijing, brain indexed to Washington — which means its true cost is not the bill of materials of Chapter 2 but that figure plus a geopolitical risk premium neither chapter could put a clean number on.

That premium is the hidden cost this chapter adds to the ledger. Some of it is paid in tariffs and licensing delays today. More of it will be paid later, and physically: every plan to "de-risk" the supply chain means standing up rare-earth refining, magnet-making, and chip fabrication on home soil — among the most energy-intensive and waste-heavy industries there are, the environmental bill for which arrives in the closing chapters of this book. For now, the components clear customs and the robots get built. The next question is who actually puts them to work — and the answer, it turns out, divides as sharply by generation as the supply chain divides by border.

6. Adoption & the Generational Divide

Chapter 5 left the components clearing customs and the robots getting built. The question this chapter asks is the one every supply chain eventually runs into: who actually buys what comes off the line, and puts it to work? The popular answer is a neat generational story — that each age cohort relates to robots in its own tidy way, and disposes of them accordingly. That story is almost entirely invented. The real generational divide is messier, better documented, and more consequential, and it is built from two forces that do not point the same way: an aging workforce that pulls robots in whether anyone is enthusiastic or not, and a younger population that uses the technology heavily yet remains wary of it.

6.1 The demographic engine

The strongest force behind robot adoption is not excitement. It is arithmetic. As a population ages and the supply of working-age people shrinks, employers automate to cover the gap — a relationship the economists Daron Acemoglu and Pascual Restrepo documented as a robust, cross-country association between aging and robot adoption: the societies running short of middle-aged workers are precisely the ones that install the most robots (Acemoglu & Restrepo, 2022). Demographics, not sentiment, set the baseline. Japan is the clearest case — its labor shortage hit a record in 2024, and automation has become the country's structural answer to a workforce that is simply running out of people (International Federation of Robotics, 2024b).

The map of where robots already work follows that logic almost exactly.

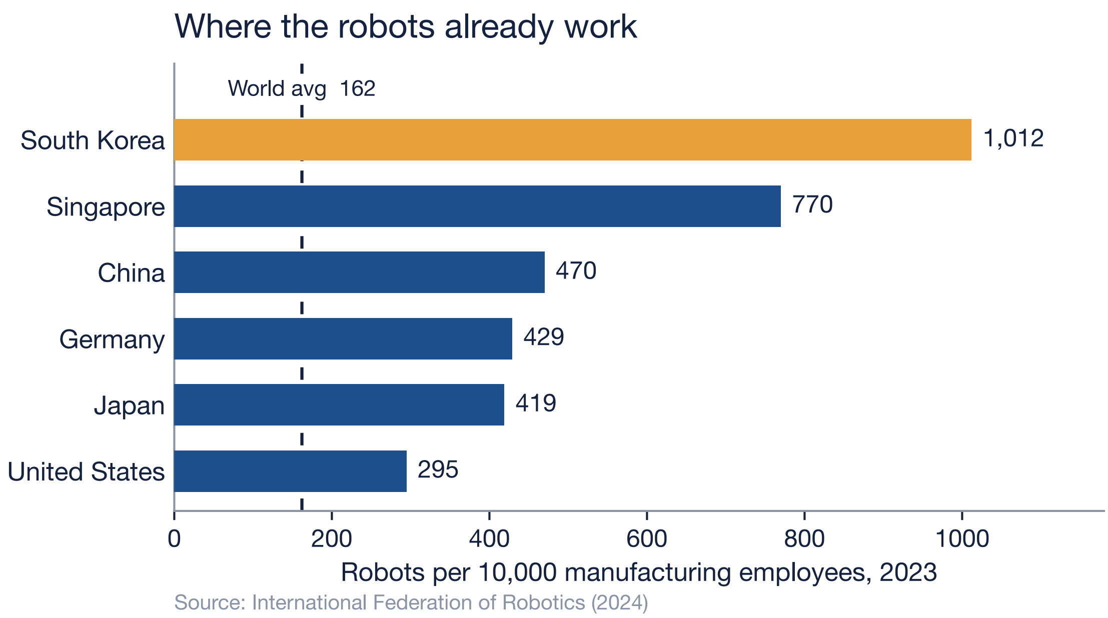

Figure 6.1: Manufacturing robot density — installed industrial robots per 10,000 employees, 2023. The leaders are the economies with the oldest or scarcest workforces: South Korea (one of the world's lowest fertility rates), Japan and Germany (aging industrial bases), and labor-tight Singapore. China has more than doubled its density since 2019 and now stands at 470 — above the United States at 295. The dashed line is the world average of 162, itself double the 2016 figure. Source: International Federation of Robotics (2024).

Read the ranking and the demographic engine is unmistakable. South Korea, with one of the world's lowest fertility rates, runs more than 1,000 robots per 10,000 factory workers — six times the world average. Singapore, Japan, and Germany follow, each an economy where labor is old, expensive, or both. The United States, with a younger population and a deeper pool of workers, sits at 295 — and has now been passed by China, whose density doubled in four years on the back of state policy, the electric-vehicle supply chain of Chapter 5, and an aging population of its own. Where the workforce is scarce, the robots are already there; where it is not, adoption lags. That is the generational divide that actually moves units, and it is a divide between national age structures, not between consumer attitudes.

6.2 The attitudinal divide

The second force is softer and runs the other way. Surveyed about how artificial intelligence makes them feel, Americans split by age — but not in the direction the marketing decks assume. Younger adults are the heavy users: about 38% of those aged 18–29 have used ChatGPT at work, against 30% of those 30–49 and 18% of those 50 and older, and roughly half of all adults under 50 now interact with some AI daily or more (Pew Research Center, 2026). Exposure falls steadily with age. Yet exposure is not endorsement. Across the whole adult population about half say AI's growing role makes them more concerned than excited, and only around one in ten lean the other way; younger adults are *slightly* more excited than their elders, but majorities of them are also wary of what AI does to human creativity and to the way people form relationships (Pew Research Center, 2026).

The honest reading is that the young do not "embrace" robots while the old "resist" them. The young adopt the software fluently and worry about it specifically; the old use it less and distrust it more broadly. Acceptance and use are two different axes, and a generation can sit high on one and low on the other. Any adoption forecast that treats youth as a simple proxy for enthusiasm is reading a curve that the data does not contain.

6.3 The matrix that isn't there

It is worth naming what this chapter does not do, because the temptation is everywhere in the industry's own materials. A tidy table circulates in robotics decks and draft reports: Baby Boomers cling to repair, Millennials replace out of "ecological guilt," Gen Z boycotts unsustainable brands, each cohort assigned a crisp disposal behavior. It is a seductive framework, and it is unsupported. Those behavioral claims trace back to no survey, dataset, or study that can be located; they are assertions repeated until they sound like findings. Under this report's standard — a claim is either sourced or it is silent — the matrix cannot be used, and is not. What can be said about generations and robots is what the demographic and survey data above actually support, and no more. The cost of intellectual honesty here is a less colorful table; the benefit is that nothing in this chapter will collapse when someone asks for the source.

6.4 Where the forces meet

Put the two forces together and the geography of adoption resolves. Robots arrive fastest where the demographic pressure is sharpest: the aging economies of East Asia, where a shrinking workforce makes automation urgent and state policy pushes it hard. Whether those populations are also more *culturally* at ease with robots is plausible but, on the evidence assembled here, unproven — the attitudinal survey data is American, and should not be assumed to describe Seoul or Shenzhen. Adoption slows where the demographic pressure eases — where the workforce is still young enough to fill the jobs itself — and the United States, younger and (by its own survey data) ambivalent about the technology, is structurally a slower adopter than its rivalry with China implies. That is a gap the next chapters in this race will keep returning to.

There is a sting in the tail, and it sets up what follows. The economies adopting humanoids fastest are also the ones that will replace them fastest — not because of any generational temperament, but because the hardware itself is about to start iterating at the speed of consumer electronics, and a machine made obsolete by next year's model is discarded whether its owner is sentimental about it or not. The generation of robots now being bought is already scheduled for obsolescence. The next chapter follows that acceleration, and the chapters after it follow what the discarded machines leave behind.

7. The Fast Hardware

Chapter 6 ended on a sting: the generation of robots now being bought is already scheduled for obsolescence — not because of any buyer's temperament, but because the hardware itself iterates at the speed of consumer electronics. This chapter follows that acceleration, and to do so it has to take a humanoid apart into the two machines it really is. There is a body — a frame, actuators, sensors, a battery — engineered the way industrial machinery has always been engineered, to last. And bolted into that body is a brain — the onboard AI compute and the model it runs — built on the opposite premise, to be surpassed within a year. The two halves keep different time. The argument of this chapter is that when you fasten them together, the faster clock wins, and it drags the whole machine's useful life down with it.

7.1 The brain's clock

The pace of the brain is not set by the robot at all. It is set by the frontier of artificial intelligence, and that frontier is moving at a rate with few precedents in industrial history. Epoch AI, the research group that specializes in measuring the compute behind frontier AI, finds that the training compute behind frontier models has grown roughly 5.3 times per year between 2010 and mid-2024 — and even after a deceleration around 2018, still climbs about 4.2 times a year, which is to say the compute defining the state of the art roughly *doubles every five to seven months* (Epoch AI, 2024). A robot's intelligence does not have to degrade for it to fall behind; the target it is measured against simply recedes that fast. That figure describes the pace of the frontier, not the decay of any single chip — what ties it to a particular robot is the release cadence of the hardware built to chase it.

And the hardware built to chase that frontier keeps the same relentless beat. At Computex 2024, NVIDIA's chief executive Jensen Huang described the company's release schedule in plain terms: "Our company has a one-year rhythm. Our basic philosophy is very simple: build the entire data center scale, disaggregate and sell to you parts on a one-year rhythm, and push everything to technology limits" (NVIDIA, 2024a). The roadmap he laid out — Hopper, Blackwell, Rubin, and the refreshed parts between them — puts a faster accelerator on the market every year. A humanoid's "brain" is built from this same lineage of AI silicon — a smaller, embedded cousin of the data-center part — paired with a foundation model of the kind NVIDIA itself launched for robots in 2024 (NVIDIA, 2024b). Both halves of that brain — the chip and the model — are designed to be eclipsed on a twelve-month cadence. The compute does not wear out. It gets outclassed.

7.2 The body's clock

The body runs on the opposite premise. A humanoid is, mechanically, a dense assembly of precision actuators — Chapter 2 put them at more than half of the entire bill of materials — built to the tolerances of industrial machinery (Bank of America Institute, 2026). And industrial machinery is engineered to last: the arms it descends from are advertised with mean time between failures as high as 100,000 operating hours, and the real-world factory data from Chapter 4 show that when a robot cell does stop, the arm itself is almost never the thing that broke (The Robot Report, 2019). A well-maintained industrial robot earns its keep over a decade or more. That is the design intent written into the body of a humanoid: years of service, slow depreciation, mechanical endurance.

So the machine carries two clocks that disagree by an order of magnitude. The body is built for roughly a decade; the brain is built to be lapped in about a year. Nothing in engineering forces those two numbers to converge — and that is precisely the problem.

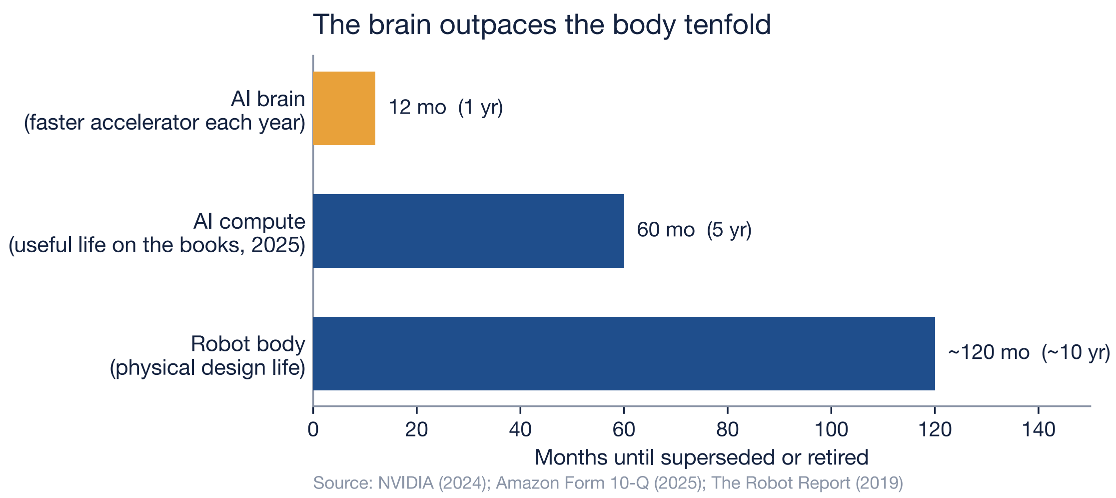

Figure 7.1: Two clocks in one machine. The brain — the onboard AI compute — is overtaken by a faster accelerator every year on NVIDIA's stated one-year rhythm (amber). The data-center industry's own books now carry that class of compute at a five-year useful life (Amazon's 2025 estimate). The physical body is engineered for roughly a decade — the design life of the industrial arms it descends from (~100,000 operating hours, advertised, ≈ ten years at near-continuous use). The three bars are not the same kind of measure — an interval until a faster part ships, an accounting useful life, and a physical design life — but laid on one axis they show the gap this chapter is about: the shortest clock decides when the whole machine is thrown away. Sources: NVIDIA (2024); Amazon Form 10-Q (2025); industrial-robot design life via The Robot Report (2019).

7.3 Two clocks, one machine

Read the chart and the pressure is plain: the shortest clock sets the ceiling on the rest. Whether any particular machine is retired early is, of course, an economic decision — the same capital-conservative buyers who run an industrial arm for a decade do not scrap working equipment on a whim, which is exactly why the body lasts as long as it does. But the chapter's claim is narrower and harder to dodge: the question governing that decision shifts off the body and onto the brain. It stops being "does the machine still work" — for fast hardware it almost always does — and becomes "is the intelligence inside it still worth deploying against a frontier that has moved." When a newer model does a job the older one cannot, the decade of mechanical life still left in the older body is stranded rather than redeemed. That is what "fast hardware" means in a humanoid: the speed does not belong to the body — the slow, durable, expensive half — but to the brain bolted inside it, and it is the brain's obsolescence, not the body's wear, that increasingly decides when the unit is replaced.

That is a different and more aggressive failure mode than the one Chapter 4 examined. There the question was whether the machine *works*; here the machine can work perfectly and still be retired, because something better exists. Reliability sets a floor under a robot's life. Obsolescence sets a ceiling — and for fast hardware the ceiling is far lower than the floor.

7.4 The market is already shortening the clock

This is not a forecast. It is already visible in the accounts of the companies that own the most AI compute on earth. For years the hyperscalers stretched the assumed life of their hardware: Microsoft, after a review in July 2022, lengthened the useful life of its server and network equipment from four years to six, citing "investments in software that increased efficiencies … as well as advances in technology" (Microsoft Corporation, 2023). Amazon followed the same logic, extending its servers from five years to six effective January 2024 — a change worth about $3.2 billion in reduced depreciation and $2.5 billion in added net income that year (Amazon.com, Inc., 2025b). The bet was that the hardware would stay useful *longer*.

Then the bet reversed. Effective January 1, 2025, Amazon shortened the useful life of a subset of its servers and networking equipment back from six years to five — and stated the reason in its own filing with the Securities and Exchange Commission.

Table 7.1: How the world's largest cloud operators have re-estimated the working life of AI compute. The 2025 reversal is the first time in this sequence the trend bends back toward a *shorter* life — and the reason given is AI itself. Sources: Microsoft Form 10-K (2023); Amazon Form 10-K (2024) and Form 10-Q (2025).

Date effective	Company	Server useful life	Stated reason
Fiscal year 2023	Microsoft	4 ⟶ 6 years	efficiency gains and "advances in technology"
Jan 1, 2024	Amazon	5 ⟶ 6 years	longer life assumed (≈$3.2 B less depreciation that year)
Jan 1, 2025	**Amazon**	**6 ⟶ 5 years**	**"the increased pace of technology development, particularly in the area of artificial intelligence and machine learning"**

The reversal cost Amazon $217 million in additional depreciation in the first quarter of 2025 alone (Amazon.com, Inc., 2025a). The number matters less than the admission behind it. The most sophisticated owners of AI compute in the world have just told their auditors that this hardware lasts *less* time than they thought a year earlier, and they have named the cause: the pace of AI. These are data-center servers, not humanoids — but a humanoid's brain is built from the same restless lineage of AI silicon, and it inherits the same shortening clock. When Amazon writes down the working life of an accelerator, it is describing the brain that will sit inside the next warehouse robot.

7.5 The escape that isn't quite an escape

The obvious engineering answer is modularity: make the brain a swappable card, upgrade the compute on its fast cycle, and keep the slow, durable body for its full decade. It is a real mitigation, and an honest account has to grant it. A genuinely modular compute module would decouple the two clocks for the part that is swapped, and it is the obvious thing for a designer to want.

But it decouples less than it promises, for two reasons. First, the brain is not only the compute card. A frontier model demands matching sensors, actuator bandwidth, and power, and Chapter 4 showed the binding limit is already physical — today's humanoids run roughly two hours on a charge (Bain & Company, 2025). A body specified in 2026 may simply not be able to host the sensing and power budget a 2030 model assumes, no matter what card you slot into it; the halves co-evolve, and the body can fall behind on its own terms. Second, even where the swap works cleanly, it does not make the obsolescence disappear — it relocates it. The superseded compute still leaves the machine and still enters the waste stream, just as a module rather than a whole robot. Modularity changes the *shape* of the discard. It does not cancel it.

The generation of humanoids now being built, then, will be outclassed long before it is worn out — and the market that owns their kind of brain is already shortening the life it assigns to them. That leaves a question this chapter cannot answer and the next one must: what becomes of a decade's worth of barely-used bodies, and of the fast-obsoleting brains stripped out and thrown after them? The machines are about to start coming off the line faster than they wear out. The chapters that follow track where they go when they do — into the e-waste wave.

8. The E-Waste Wave

Chapter 7 left a generation of humanoids outclassed long before they are worn out, their bodies stranded and their brains stripped out and thrown after them. This chapter asks the question the lifecycle finally forces: where does all of that actually go? The answer is not a clean recycling loop. It is a waste stream that the world is already losing control of — and the robots are arriving into it at the worst possible moment, carrying the worst possible materials.

8.1 A system already losing the race

Begin with the ground the robots are landing on. The world generated 62 million tonnes of electronic waste in 2022, and the UN's Global E-waste Monitor projects that figure will reach 82 million tonnes by 2030, rising by about 2.6 million tonnes every year (Baldé et al., 2024). That alone would be manageable if recycling kept pace. It does not. Less than a quarter of that mass — 22.3% in 2022 — was documented as formally collected and recycled, and the monitor projects the rate will *fall* to 20% by 2030 (Baldé et al., 2024). E-waste is growing roughly five times faster than the recycling meant to absorb it. The system is not approaching the problem; it is falling further behind it every year.

8.2 The value we bury

The loss is not only environmental. It is financial, and the numbers are large enough to matter to the argument of this book. The e-waste generated in 2022 contained metals worth about $91 billion — copper, gold, iron, the whole periodic table of a modern economy. Of that, only around $28 billion was recovered. The rest, some $62 billion in recoverable resources, was simply unaccounted for: landfilled, incinerated, or scattered through informal dumps where the materials are lost (Baldé et al., 2024).

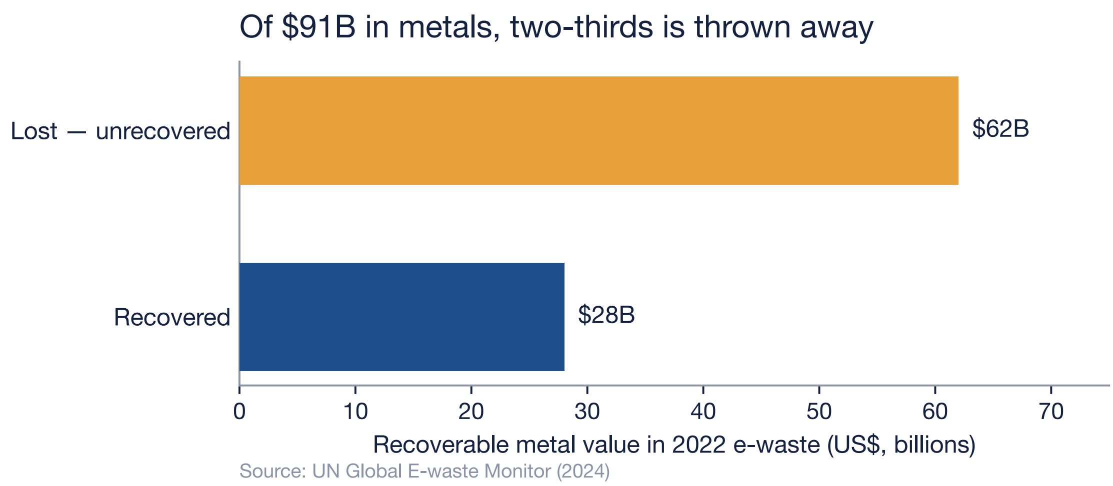

Figure 8.1: The metals embedded in the world's 2022 e-waste were worth about US$91 billion. Only ~$28 billion was recovered through recycling; the remaining ~$62 billion in recoverable resources went unaccounted for — landfilled, burned, or lost to informal dumping (amber). The figure is a measured year, not a forecast. Source: UN Global E-waste Monitor (2024).

And the single worst-recovered material is the one this book has spent two chapters on. Of all the rare-earth elements the world demands every year, only about **1%** is supplied by e-waste recycling (Baldé et al., 2024). Recall what that means against Chapter 5: the United States and China built export controls, tariffs, and stockpiles around rare-earth magnets because China refines roughly 91% of the world's supply and makes about 94% of the finished magnets (International Energy Agency, 2025). Those same magnets, at the end of a machine's life, are recovered at a rate of one percent. The world fights for the material at the mine and throws it away at the dump.

8.3 Why a humanoid is the worst kind of e-waste

Into that failing system, now feed a humanoid. A smartphone weighs under 0.2 kilograms; a humanoid robot weighs on the order of 50 — reported figures for Tesla's Optimus run from roughly 47 to 73 kilograms across its generations, so a single robot is hundreds of times the mass of the phone it is so often compared to.

Table 8.1: A humanoid is not a big phone. It concentrates exactly the materials the recycling system handles worst. Sources: e-waste stream and recycling figures, UN Global E-waste Monitor (2024); actuator share of the bill of materials, Chapter 2 (Bank of America Institute, 2026); Optimus mass is a reported range, as Tesla publishes no single official figure.

	A smartphone	A humanoid robot
Typical mass	under 0.2 kg	~50 kg — hundreds of times a phone
Rare-earth content	a trace, in speakers and motors	a permanent magnet in *every* actuator — and actuators are more than half the bill of materials
Energy store	one small cell	a full-body battery pack
End-of-life path	part of the 62 Mt stream; ~22% recycled	falls into that same stream, with no recovery channel of its own

But mass is the least of it, and it would be dishonest to pretend otherwise: even millions of humanoids would be a rounding error against 82 million tonnes a year. The danger is not the weight — it is the composition. A humanoid concentrates precisely the materials the system recovers worst. Nearly every electric actuator that moves it is built around a neodymium-iron-boron magnet, and Chapter 2 established that actuators are more than half of a humanoid's entire bill of materials (Bank of America Institute, 2026); bonded into dense, mixed-metal assemblies, those magnets are exactly what the 1% recovery rate leaves behind. Each discarded humanoid is therefore a small, heavy parcel of the contested chokepoint material, posted directly into the 78% of e-waste that is never formally recovered.

Here honesty requires naming a gap rather than papering over it. Robots are not yet broken out as a category in global e-waste statistics; there is no measured tonnage of "robot e-waste," and the speculative billion-units-by-2050 projections that circulate in industry reports rest on assumptions their authors cannot source. So this chapter does not forecast a tonnage. What it can say, on the evidence, is structural and harder to dismiss: the obsolescence clock of Chapter 7 is set to feed rising volumes of some of the densest, most critical-material-laden, and least-recyclable hardware in the stream into a system whose recovery rate is already falling.

8.4 The wave meets the chokepoint

There is a closing irony the rest of this book has been building toward. The same nations racing to secure rare earths — restricting their export, tariffing their import, subsidizing their mining — are simultaneously building the machines that will bury those rare earths at a 1% recovery rate. The robot race optimizes furiously for *getting* the chokepoint material and barely at all for *keeping* it. Every chapter so far has measured a cost on the way up the lifecycle; this is the cost at the bottom, where the material the whole race was fought over ends up scattered through landfills.

The obvious rejoinder is that this is precisely what a circular economy is supposed to fix — and, fairly stated, a humanoid is a better candidate for recovery than almost anything in the e-waste stream: it is high-value, individually identifiable, and owned in fleets by businesses rather than forgotten in a billion household drawers. In principle it should be the easy case. But "in principle" is carrying a great deal of weight in that sentence, and the measured trend points the other way — the recovery rate is falling, not rising, and the channels that might keep a retired robot's materials in circulation are still improvised rather than systematic. Whether that loop can actually be closed, and what it would cost to close it, is the question the next chapter takes up.

9. The Circular Economy Response

Chapter 8 ended on a deferral: a humanoid is, in principle, the easy case for recovery — high-value, individually identifiable, owned in fleets — and yet the measured trend runs the other way, with the recovery rate falling rather than rising. That gap between what should happen and what does is the subject of this chapter. The circular economy is not a hypothetical here. For one kind of robot it already exists, it works, and it has the warranties and the carbon figures to prove it. The question is whether that working loop is the answer to the humanoid problem — or a model that quietly breaks the moment a humanoid enters it.

9.1 A loop that already turns

Start with the good news, because it is real. Walk into ABB's robotics business and you find a circular economy in full operation. A customer with an aging industrial arm does not have to scrap it; ABB will buy it back, strip it down at one of its Global Remanufacture and Workshop Repair Centers — in Czechia, Michigan, Shanghai, Brazil, Mexico, Germany, Vietnam — and rebuild it from the company's original design plans and dimensional data. Every unit passes a detailed inspection and a minimum sixteen-hour functioning test before it can be labelled a certified remanufactured robot, and it ships with a warranty — at least twelve months, and two years in ABB's program announcement — plus the same installation, training, and service support as a new machine (ABB, 2020; 2026). ABB's own numbers for the process are striking: refurbishing a robot produces **75% less CO_2** than building a new one, and **60–80% of the robot can be reused**, with the remainder sent to certified recycling partners (ABB, 2026). By 2020, ABB had put thousands of robots through this loop across twenty-five years of operation (ABB, 2020).

And the channel is not exotic. ABB's buy-back leg takes in legacy and inactive robots that would otherwise be scrapped and feeds them back into a genuine secondary market in certified used arms (ABB, 2020). The same instinct operates at hyperscale on the compute side of the ledger: in 2024 Google collected roughly **8.8 million** components from decommissioned hardware to reuse or resell through its reverse-supply-chain program, drew **44%** of the components used to build and maintain its servers from reused inventory, and diverted **84%** of its data-center operational waste from landfill (Google, 2025). The circular economy, in other words, is not waiting to be invented. Where the hardware is valuable, standardized, and owned by organizations that keep records, the loop closes.

9.2 Real — and a rounding error

The trouble is scale, and it is worth being precise about. The world installed about **542,000** new industrial robots in 2024 and now runs an operational stock of roughly **4.66 million** machines (International Federation of Robotics, 2025b). Against that flow, "thousands of robots over twenty-five years" — even read generously as a few hundred a year, and even multiplied across every OEM running a similar program — works out, on a back-of-the-envelope estimate, to a fraction of one percent of what is installed annually. The loop turns; it does not turn fast enough to absorb the stream feeding into it. This is not a hypothetical limit. It is exactly what Chapter 8 measured from the other end: if remanufacturing and recycling were keeping pace, the broad e-waste recovery rate those retired machines fall into would be climbing, not falling toward 20%.

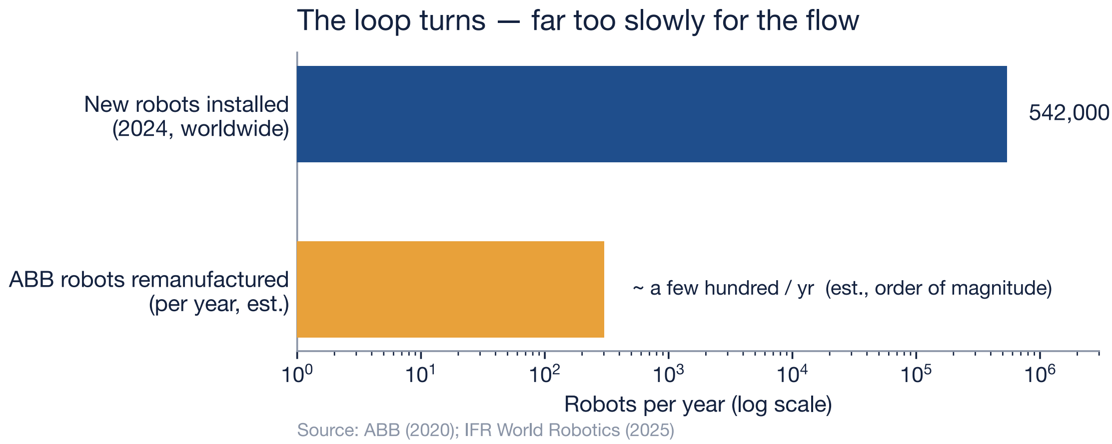

Figure 9.1: The circular channel is real but small against the flow it must absorb. The world installs on the order of 542,000 new industrial robots a year (blue, IFR World Robotics 2025). The best-documented remanufacturing program — ABB's — has refurbished 'thousands of robots over twenty-five years,' on the order of a few hundred a year (amber). ABB publishes no exact annual figure and is one OEM, not the whole industry; the bar is plotted as an order of magnitude on a log scale, where the point is the roughly thousand-fold gap, not a precise count. Even generous industry-wide multiples stay well under 1% of annual installs. Sources: ABB (2020); IFR World Robotics (2025).

9.3 Why the loop does not transfer to the humanoid

Scale is the smaller objection. The larger one is that the loop which closes so cleanly for an industrial arm is built on a property the humanoid does not share. An industrial robot's value lives in its mechanics — the gearboxes, the motors, the rigid structure — and mechanics do not go out of date. A 2010 welding arm rebuilt to original specification welds exactly as well as it did when new, which is why a warranty on a remanufactured unit is an honest promise. The humanoid inverts this. Its value lives in its intelligence, and Chapter 7 established that the intelligence is lapped roughly every twelve months while the body is engineered to last a decade. By the time a humanoid's body is worn enough to be worth remanufacturing, its brain — and very likely its entire software platform — is long obsolete. Remanufacturing would faithfully restore the part of the machine that still works and could do nothing for the part that decides whether anyone wants it.

Table 9.1: The circular loop closes for the industrial arm because its value is mechanical and its mechanics endure. A humanoid breaks every assumption the loop depends on. Sources: remanufacturing process and reuse share, ABB (ABB, 2020; 2026); obsolescence horizon, Chapter 7; actuator share of the bill of materials, Chapter 2 (Bank of America Institute, 2026).

Dimension	Industrial robot arm	Humanoid robot
Where the value lives	In the mechanics — gearboxes, motors, structure	In the intelligence — the compute and the model
Obsolescence horizon	A decade or more; mechanics do not date	Brain lapped in ~12 months (Ch. 7); body outlives its usefulness
What remanufacturing restores	The valuable part — the mechanics	Only the durable part — the body — not the valuable part
Standardization	Stable, long-lived models; parts interchange	Bespoke, fast-iterating platforms; little cross-generation reuse
Critical-material recovery	NdFeB magnets in a handful of joints	A magnet in *every* actuator; actuators are >½ the BOM (Ch. 2)

And then there is the material itself. ABB's 60–80% reuse figure is a share of the robot by parts and mass — the heavy, generic, reusable structure. The neodymium-iron-boron magnet inside each actuator is precisely what falls into the other 20–40%, the portion "sent to certified recycling partners." Chapter 8 already named what happens there: of all the rare-earth element demand in the world, recycling supplies about 1% (Baldé et al., 2024). The two figures describe different denominators and cannot be added, but their direction is the same and it is the crux of this book — the circular economy, working at its best on an industrial robot, recovers the common metals and still loses the contested ones. A humanoid carries a permanent magnet in *every* actuator, and actuators are more than half its bill of materials (Chapter 2) — so it walks more of that chokepoint material into the 1% recovery zone than any machine the loop was designed around.

9.4 What closing it would actually take

None of this makes a humanoid circular economy impossible. It makes clear what it would have to be, and how far that is from what exists. The single change that would matter most follows directly from the obsolescence clock: design the brain to be swapped, not buried. A modular compute module that can be upgraded while the durable body stays in service would let the part that lasts a decade actually last a decade, instead of being scrapped because the part that lasts a year is soldered to it. Around that, the familiar circular-economy machinery — design for disassembly so the magnets come out cleanly, and extended-producer-responsibility take-back rules so the fleets are routed to qualified battery and metallurgical recyclers rather than the informal stream — would do the rest of the work. These are not technologies awaiting discovery; the industrial-arm program proves the logistics are solvable. They are choices, and the measured trend says the market is not making them on its own: left to price signals, the recovery rate is falling, not rising.

That is the honest answer to the question Chapter 8 deferred. The loop can be closed, but not by assuming the humanoid will inherit the industrial robot's circular economy — because the humanoid breaks the very feature that makes that economy pay. Closing it requires building something the industry has not yet built, against a clock that is currently winning. Which leaves a final accounting to be done: with every stage of the lifecycle now measured — the price, the capital, the reliability, the trade chokepoints, the adoption, the obsolescence, the waste, and now the partial and conditional recovery — the last chapter draws the full balance and asks what the AI robot race actually costs, net of everything we have managed to claw back.

10. The Net Reckoning & Solutions

The book opened on a duel — the United States betting on the brain, China on the body, each convinced its theory of the race would win (Bank of America Institute, 2026; Figure AI, 2025). That framing asks a single question: who finishes first. Nine chapters later it is the wrong question, or at least an incomplete one, because the race is no longer hypothetical and neither is its bill. The price of the machine has been measured, and so have the capital behind it, the reliability it lacks, the trade chokepoints it depends on, the way it is adopted, the clock that obsoletes it, the waste it becomes, and the little of that waste anyone recovers. What is left is the arithmetic the title of this book has pointed at from the first page — not *will the robots come*, because they are coming, but *what does the race actually cost, net of everything we have managed to claw back*. This chapter draws that balance.

10.1 The balance sheet

Set every stage of the lifecycle in one column and a pattern appears that no single chapter could show on its own. Each row below is a figure already verified in its own chapter; the last column asks the only question that matters for a reckoning — at this stage, does the race pour in effort, or look away?

Table 10.1: Every measured stage of the humanoid's life, and where the race spends its attention. The split is not subtle: the four acquisition stages are optimized, the four keeping stages are neglected or only conditionally addressed. Sources by row — build (Bank of America Institute, 2026); fund (Crunchbase News, 2025; Figure AI, 2025; International Federation of Robotics, 2025a); source (Congressional Research Service, 2025; International Energy Agency, 2025; Office of the United States Trade Representative, 2024); deploy (Acemoglu & Restrepo, 2022; International Federation of Robotics, 2024a); run (Bain & Company, 2025); outdate (Amazon.com, Inc., 2025a; NVIDIA, 2024a; The Robot Report, 2019); discard (Baldé et al., 2024); recover (ABB, 2020; 2026; International Federation of Robotics, 2025b).

Lifecycle stage	What the book measured	Does the race optimize for it — or neglect it?
Build — the bill of materials (Ch. 2)	China-built BOM ≈ \$35,000 in 2025, engineered to fall below \$17,000 by 2030; actuators alone are more than half of it	**Optimized hard.** Unit cost is being driven down deliberately, year over year.
Fund — the capital (Ch. 3)	A ¥1 trillion (~\$138B) Chinese state fund; Figure AI valued at \$39B; robotics venture funding at a multiyear high	**Optimized hard.** Money is the least scarce input in the whole race.
Source — the materials (Ch. 5)	China refines ~91% of rare earths and makes ~94% of the magnets; the US controls the frontier chips and now taxes the magnets it barely makes at 25%	**Fought over hard — at the front.** Acquisition is open national strategy; the recovery end is nobody's.
Deploy — adoption (Ch. 6)	Robot density of 1,012 in Korea and 470 in China against a world average of 162, pulled by aging workforces	**Optimized.** Installation is racing ahead of any evidence on how the machines hold up.
Run — reliability (Ch. 4)	No humanoid MTBF has ever been disclosed; today's machines run roughly two hours on a charge	**Neglected.** No reliability or total-cost-of-ownership data exists at scale.
Outdate — the clock (Ch. 7)	The brain is lapped about every twelve months while the body is built to last a decade; Amazon shortened its own server life and named AI as the reason	**Neglected — and built in.** The mismatch manufactures waste by design.
Discard — the waste (Ch. 8)	Global e-waste rising from 62 to 82 Mt; ~\$62B in recoverable metals left unrecovered in a single year's e-waste; ~1% of rare-earth demand met by recycling	**Neglected.** Recovery is *losing* ground as the volume climbs.
Recover — the loop (Ch. 9)	ABB remanufacturing cuts CO_2 75% and reuses 60–80% of an arm — real, but a rounding error against 4.66M robots in service, and it does not transfer to the humanoid	**Conditional.** Works for the durable arm; fails the disposable brain.

Read the table from top to bottom and the shape is unmistakable. The first four stages — building the machine, funding it, sourcing its materials, putting it to work — are the *acquisition* half of the lifecycle, and every one of them is optimized furiously. A bill of materials is being halved on purpose; a trillion-yuan fund stands behind it; the magnets are a matter of national security on both sides of the Pacific; the robots are being installed faster than anyone can say whether they last. The bottom four — running the machine, watching it fall behind, discarding it, recovering what is left — are the *keeping* half, and every one of them is either neglected or, in the single case where a loop exists, conditional and small. The line between row four and row five is the line where the money and the geopolitical attention stop.

10.2 The net

This is the hidden cost the title promised, and the honest finding is that it is not a number — it is a gap. The race optimizes, with extraordinary intensity, for getting the machine: get the chip, get the magnet, drive down the build, pour in the capital, install the fleet. It optimizes almost not at all for keeping what that machine is made of, or even for knowing how long the machine will work. The cost is hidden precisely because it lives in the half of the lifecycle nobody is measuring — and the absence of measurement is itself the evidence. The front half of this book is denominated in precise dollars: a $35,000 bill of materials, a $39 billion valuation, a 25% tariff. The back half is denominated in absences: no disclosed MTBF, no robot-specific e-waste tonnage, a recycled rare-earth supply that meets barely 1% of demand. The asymmetry in the *data* is a faithful mirror of the asymmetry in the *effort*.

Hold two of those figures next to each other and the gap becomes a single sentence. China makes roughly 94% of the world's permanent magnets, and the United States now taxes them at 25% and treats the supply as a national-security file (International Energy Agency, 2025; Office of the United States Trade Representative, 2024) — the same rare-earth elements of which recycling supplies barely 1% of world demand once the machines that carry them are thrown away (Baldé et al., 2024). We fight for the magnet at the mine and abandon it in the dump. It is the same material, moving through the same supply chain, met with opposite urgency at its two ends. The race will back the technology with a $138 billion national hard-tech fund (International Federation of Robotics, 2025a) and require no published figure on how long the machine runs before it breaks (Bain & Company, 2025). That is the net of the AI robot race as the evidence now stands: a machine the world is racing to acquire and has barely begun to plan to keep.

10.3 What it would take

The reckoning would be unbearable if it were also a dead end, and it is not. Chapter 9 showed that the logistics of recovery are already solved for one kind of robot — ABB's remanufacturing loop has the warranties and the carbon numbers to prove it (ABB, 2020; 2026). The work that remains is to make the humanoid fit a loop instead of breaking it, and the table above says where to start: the neglected back half, in order of leverage.

The single highest-leverage change follows directly from the obsolescence clock. *Design the brain to be swapped, not buried* — a modular, upgradeable compute unit so the part that is lapped every twelve months can be replaced while the body that was built for a decade actually serves a decade, instead of being scrapped because a year-old chip is soldered to it. This is a design proposal, not a measured fact, but it is the one intervention that attacks the specific failure Chapter 7 identified rather than working around it. Around that, the familiar circular-economy machinery would do the rest: *design for disassembly* so the magnet in every actuator can be pulled out cleanly instead of shredded with the chassis, and *extended-producer-responsibility take-back rules* so fleets are routed to qualified battery and metallurgical recyclers rather than the informal e-waste stream that already swallows most of what we discard (Baldé et al., 2024). None of these awaits a technological breakthrough; the industrial-arm program proves the logistics are tractable (ABB, 2026). They await a decision, and the measured trend says the market is not making it on price signals alone — left to itself, the recovery rate is falling, not rising.

There is a geopolitical reading of this that should change how the back half is valued. If you cannot out-mine the country that controls 94% of the magnets, then lifting recycling above the ~1% of rare-earth demand it now supplies stops being an environmental nicety and becomes industrial strategy — the cheapest domestic rare-earth "mine" a chip-controlling, magnet-poor nation has is the fleet of machines it is already throwing away. Securing the chokepoint at the front and recovering the material at the back are not two policies; they are the same policy seen from its two ends, and at present only one end is funded. The reckoning, in other words, points at its own solution: the gap between acquisition and keeping is not a law of nature but a choice that is currently being made by default.

The accounting is now complete. Every stage of the machine's life has a figure or a documented absence beside it, and the balance is drawn — a race optimized for the front half and blind to the back. What the accounting cannot hold — what it means to build a machine in our own image and design its obsolescence into the same blueprint — is where the conclusion turns.

11. Conclusion: In Our Own Image

In Pixar's *WALL·E* (2008), the Earth has been abandoned. The people have gone to live in the sky, and what they left behind is a planet buried under its own discarded things — towers of compacted trash standing where skyscrapers used to, a horizon made of waste. One small robot is still there, still doing the job it was built for: gathering the refuse, pressing it into neat cubes, stacking them into the skyline. He keeps himself running by scavenging parts from the bodies of other robots like him that have already stopped. It is a children's film. It is also the most exact picture anyone has drawn of the destination this book spent ten chapters measuring the road toward.

The film is fiction, and nothing in the preceding pages forecasts that ending. But the parallel is uncomfortably tight, because the robot race turns the same screw one more time. *WALL·E* is a story about a machine left to clean up after people who threw everything away. This book is about something stranger: we are now building machines in our own image — humanoids, with our shape and our hands — on a blueprint whose own logic makes them disposable. Chapter 7 showed why. The brain is engineered to be outclassed in roughly a year while the body is built to last a decade, so the most human-looking machine we have ever made arrives with its own obsolescence already drawn into the plan. We are not only manufacturing the trash a future salvager would compact. We are manufacturing the salvager, too.

Strip away the figures and the finding underneath them is simple, and it is the hidden cost of the title. The race is extraordinary at the first half of the story and very nearly blind to the second. It optimizes furiously for *getting* the machine — driving a bill of materials from $35,000 down toward $17,000 in a handful of years (Bank of America Institute, 2026), pouring in state funds and venture billions, fighting nation against nation over the magnets and the chips — and it optimizes for almost nothing about *keeping* it: not the reliability no manufacturer will disclose, not the material it discards at the end. The sharpest single image of that gap is one family of metals. The same rare earths the world restricts, tariffs, and subsidizes at the mine, recycling feeds back into barely one percent of what that world demands (Baldé et al., 2024). We pour national policy into securing that magnet at the mine, and then let it fall, unrecovered, out the bottom of the very machine we built it for.

It would be easy to read all of this as a prophecy, and *WALL·E* as the ending it points to. It is not, and that is the most important thing this book has to say. Chapter 10 argued that the distance between getting and keeping is not a law of nature; it is a default — the result of measuring, funding, and rewarding only the front half of a machine's life. The same engineering ingenuity that is driving a humanoid's build cost down by half in a handful of years could, turned the other way, design a brain meant to be swapped instead of buried and a magnet meant to be pulled back out. A buried Earth is one possible ending. The whole argument of these chapters is that it is a *choice* of ending, and the choice has not yet been made.

There is a reason the machine at the center of this race is shaped like us. A humanoid is a mirror by design, and what it reflects back is not only our cleverness but our priorities — which half of ourselves we are pouring into the things we build. The half that races to acquire, or the half that takes responsibility for what it makes and for what it leaves behind. *WALL·E* imagined a world that answered that question by accident and left a single small robot to live with the answer. We are still early enough to answer it on purpose.

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